

Lecture 5: Sets and Beyond

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1 From Predicates to Sets

Last class, we learned that a predicate has a specified domain and becomes true or false when we supply an input. Today, we use the inputs that make a predicate true to define a **set**.

Definition. A set is a well-defined collection of distinct objects from a domain. The objects in a set are its **elements**. Membership must be unambiguous: for each object in the domain, either it belongs to the set or it does not. This does not promise that membership is easy to check, or that an algorithm can always decide it.

We have already seen many sets:

$$\mathbb{N} = \{0, 1, 2, 3, \dots\}, \quad \mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}, \quad \mathbb{R}, \quad \text{Boolean} = \{\text{True}, \text{False}\}.$$

Here $\mathbb{R}$ is the set of real numbers. As in our earlier lectures, we include 0 in $\mathbb{N}$.

To see how a predicate describes a set, take the domain $D = \{1, 2, 3, 4, 5, 6\}$ and the predicate “ x is even.” The inputs that make this predicate true are 2, 4, and 6. We can describe their set either by the rule that selects them or by listing them:

$$E = \{x \in D : x \text{ is even}\} = \{2, 4, 6\}.$$

Describing a set using a predicate is called **set-builder notation**. In general, $\{x \in D : P(x)\}$ means “the set of all x in D for which $P(x)$ is true.” Listing $\{2, 4, 6\}$ gives the same membership information directly.

We write $x \in E$ when x is an element of E , and $x \notin E$ when it is not. Thus $4 \in E$ and $5 \notin E$. This connects sets back to predicates: for each input $x \in D$, the statement “ $x \in E$ ” has exactly the same truth value as “ x is even.” More generally, every set contained in D defines its own membership predicate on D .

Because membership determines a set, rearranging the list or repeating an entry does not change it:

$$E = \{2, 4, 6\} = \{6, 2, 4\} = \{2, 2, 4, 6\}.$$

In each description, the same three numbers belong to E . The number of distinct elements in a finite set is its **cardinality**, written $|E|$. Here $|E| = 3$, even when we write 2 more than once.

A predicate can also select no elements. On the same domain D , the predicate “ $x > 6$ ” is false for every input. Hence $\{x \in D : x > 6\} = \emptyset$, the **empty set**, also written $\{\}$. It contains no elements, so $|\emptyset| = 0$.

Our domain need not contain only numbers. If it allows both numbers and sets, we can also form

$$T = \{2, \{4, 6\}\}.$$

This is a different set from E : its two elements are the number 2 and the set $\{4, 6\}$. Thus $|T| = 2$ and $\{4, 6\} \in T$, but $4 \notin T$. The number 4 belongs to the inner set; it is not itself an element of T .

The empty set can likewise be an element of another set. Consequently, $\{\emptyset\}$ contains one element, even though that element contains none: $|\{\emptyset\}| = 1$, whereas $|\emptyset| = 0$. When reading a set, we must keep track of exactly which objects its outermost braces contain.

2 Subsets and Equality

Subset. We write $A \subseteq B$ if every element of A also belongs to B . In the language of predicates,

$$A \subseteq B \iff \text{for every } x, (x \in A \Rightarrow x \in B).$$

For example, $\{1, 3\} \subseteq \{1, 2, 3\}$. Every set is a subset of itself. Also, $\emptyset \subseteq B$ for every set B : there is no element of the empty set that could violate the condition.

Proper subset. We write $A \subset B$ if $A \subseteq B$ and $A \neq B$. Equivalently, all elements of A belong to B , and B has at least one element that is not in A . For example, $\{1, 3\} \subset \{1, 2, 3\}$.

Equality. Two sets are equal exactly when they have the same elements:

$$A = B \iff A \subseteq B \text{ and } B \subseteq A.$$

Membership and containment are different. If $B = \{1, 2, 3\}$, then $1 \in B$ and $\{1\} \subseteq B$, but $\{1\} \notin B$. The elements of B are the numbers 1, 2, and 3, not singleton sets.

Cardinality is not equality. The sets $\{1, 2\}$ and $\{7, 8\}$ have the same cardinality, but they are not equal. For finite sets, $A \subset B$ implies $|A| < |B|$. We will see later that this conclusion can fail for infinite sets.

3 Operations on Sets

Fix a **universal set** U , the domain under discussion, and let $A, B \subseteq U$. Set operations combine their membership predicates. In each Venn diagram, the rectangle is U and shading marks the result. For the examples below, use

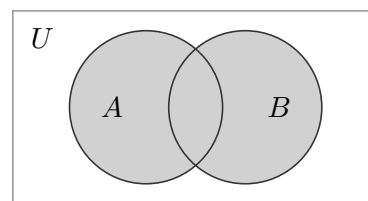
$$U = \{1, 2, 3, 4, 5\}, \quad A = \{1, 2, 3\}, \quad B = \{3, 4\}.$$

Union $A \cup B$

In A or B , including both.

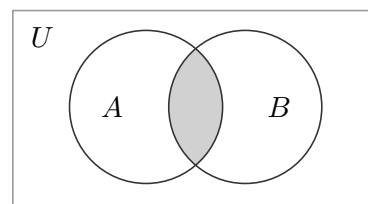
$$A \cup B = \{x \in U : x \in A \text{ or } x \in B\}$$

Example: $A \cup B = \{1, 2, 3, 4\}$.

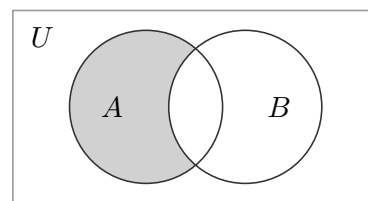


Intersection $A \cap B$ In *both* A and B .

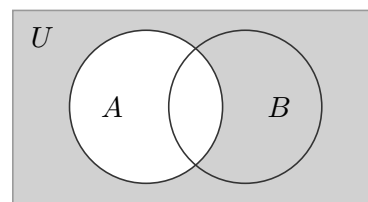
$$A \cap B = \{x \in U : x \in A \text{ and } x \in B\}$$

Example: $A \cap B = \{3\}$.**Difference** $A \setminus B$ In A but *not* in B .

$$A \setminus B = \{x \in U : x \in A \text{ and } x \notin B\}$$

Example: $A \setminus B = \{1, 2\}$.**Complement** $\bar{A}$ Outside A , within the chosen universe U .

$$\bar{A} = U \setminus A = \{x \in U : x \notin A\}$$

Example: $\bar{A} = \{4, 5\}$.

Notice that $B \setminus A = \{4\}$, so difference is not commutative. Two sets are **disjoint** when their intersection is empty.

Intuition is not a proof. Venn diagrams make overlap and exclusion visible. An informal picture alone does not prove an identity: membership must agree for every element of arbitrary sets, including empty, equal, or nested sets.

4 Set Laws and How to Prove Them

For each fixed $x \in U$, membership predicates such as $x \in A$ and $x \in B$ are Boolean statements. Union, intersection, and complement apply OR, AND, and NOT to those predicates. **Set laws therefore reduce to Boolean laws applied to the membership predicates for every element:** two sets are equal when their membership predicates agree for all $x \in U$. All complements below use this same universal set U .

Law	Identities
Commutative	$A \cup B = B \cup A; \quad A \cap B = B \cap A$
Associative	$(A \cup B) \cup C = A \cup (B \cup C)$ $(A \cap B) \cap C = A \cap (B \cap C)$
Distributive	$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
Identity	$A \cup \emptyset = A; \quad A \cap U = A$
Domination	$A \cup U = U; \quad A \cap \emptyset = \emptyset$

Law	Identities
Idempotent	$A \cup A = A; \quad A \cap A = A$
Complement	$A \cup \overline{A} = U; \quad A \cap \overline{A} = \emptyset; \quad \overline{\overline{A}} = A$
Absorption	$A \cup (A \cap B) = A; \quad A \cap (A \cup B) = A$
De Morgan	$\overline{A \cup B} = \overline{A} \cap \overline{B}$ $\overline{A \cap B} = \overline{A} \cup \overline{B}$
Difference	$A \setminus B = A \cap \overline{B}$

4.1 Three Claims Using Membership and Logic

Claim 1 (Distributivity). $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.

Proof. For any $x \in U$, distributivity for propositions gives

$$\begin{aligned}
 x \in A \cup (B \cap C) &\iff x \in A \text{ or } (x \in B \text{ and } x \in C) \\
 &\iff (x \in A \text{ or } x \in B) \text{ and } (x \in A \text{ or } x \in C) \\
 &\iff x \in (A \cup B) \cap (A \cup C).
 \end{aligned}$$

□

Claim 2 (De Morgan's law). $\overline{A \cup B} = \overline{A} \cap \overline{B}$.

Proof. For any $x \in U$, De Morgan's law for propositions gives

$$\begin{aligned}
 x \in \overline{A \cup B} &\iff \text{not } (x \in A \text{ or } x \in B) \\
 &\iff x \notin A \text{ and } x \notin B \\
 &\iff x \in \overline{A} \cap \overline{B}.
 \end{aligned}$$

□

Claim 3 (Subsets and difference). If $A \subseteq B$, then $A \setminus C \subseteq B \setminus C$.

Proof. Take any $x \in A \setminus C$. Since $A \subseteq B$, we have $x \in B$, while $x \notin C$. Thus $x \in B \setminus C$. □

To **disprove** a proposed identity, a counterexample is enough. For example, $A \setminus B = B \setminus A$ is false in general: take $A = \{1\}$ and $B = \{2\}$. The left side is $\{1\}$ and the right side is $\{2\}$.

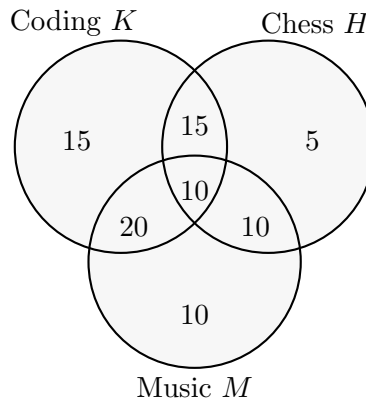
Proof Strategy. Start with an arbitrary element, unpack the definitions, and use the assumptions to establish the required membership. A Venn diagram can suggest the argument; the element-based proof explains why it works for all sets.

4.2 Example Problem: Counting Students in Clubs

Let K , H , and M be the Coding, Chess, and Music clubs. We know

$$\begin{aligned}
 |K| &= 60, & |H| &= 40, & |M| &= 50, \\
 |K \cap H| &= 25, & |H \cap M| &= 20, & |K \cap M| &= 30, \\
 |K \cap H \cap M| &= 10.
 \end{aligned}$$

How many students belong to exactly one club? Pairwise intersections include the students in all three clubs. Start with the center, then subtract 10 from each pairwise count to obtain the pair-only regions: 15, 10, and 20.



For each single-club region, subtract both pairwise overlaps and add back the triple overlap, which was subtracted twice:

$$\text{Coding only} = 60 - 25 - 30 + 10 = 15,$$

$$\text{Chess only} = 40 - 25 - 20 + 10 = 5,$$

$$\text{Music only} = 50 - 30 - 20 + 10 = 10.$$

These regions are disjoint, so $15 + 5 + 10 = \boxed{30}$ students belong to **exactly one club**. Similarly, $15 + 10 + 20 = 45$ belong to exactly two clubs, and $45 + 10 = 55$ belong to at least two.

A preview of inclusion-exclusion. This principle relates the sizes of overlapping sets, their intersections, and their union, allowing us to find missing counts. We will study it later. For three sets, add the individual sizes, subtract the pairwise intersections, and add back the triple intersection:

$$\begin{aligned} |K \cup H \cup M| &= |K| + |H| + |M| - |K \cap H| - |K \cap M| - |H \cap M| \\ &\quad + |K \cap H \cap M| \\ &= 60 + 40 + 50 - 25 - 30 - 20 + 10 = \boxed{85}. \end{aligned}$$

Thus 85 students belong to at least one club. If there are N students in total, $N - 85$ belong to none.

5 Power Sets and Bitmasks

Definition. The **power set** of A , written $\mathcal{P}(A)$, is the set of all subsets of A :

$$\mathcal{P}(A) = \{S : S \subseteq A\}.$$

For $A = \{1, 2, 3\}$,

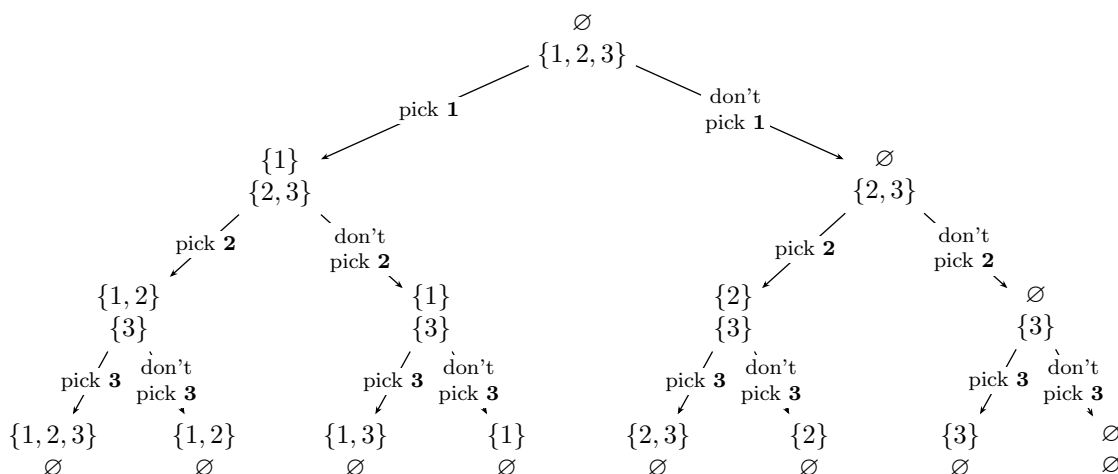
$$\mathcal{P}(A) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}.$$

Its elements are themselves sets. In particular, $\{1, 3\} \in \mathcal{P}(A)$ because $\{1, 3\} \subseteq A$. Both $\emptyset$ and A always belong to $\mathcal{P}(A)$. Also, $\mathcal{P}(\emptyset) = \{\emptyset\}$, so the power set of the empty set has one element.

Intuition. To build a subset, consider each element of A and decide whether to *pick it* or *not pick it*. For n elements, there are two choices at each of n steps, suggesting 2^n subsets. A recursive tree and a bitmask record these same decisions.

5.1 The Choices Form a Recursive Tree

Each node records the **elements picked** on its first line and the **elements still available** on its second line. For $\{1, 2, 3\}$, consider 1 first, then 2, then 3. Each branch names the decision and its element: *pick 1* or *don't pick 1*, then the corresponding choices for 2 and 3.



Each leaf has no remaining choices. Its picked set is one of the eight subsets. For example, picking 1, picking 2, and not picking 3 leads to $\{1, 2\}$.

5.2 Record the Same Choices as a Bitmask

A **bitmask** records one bit for each element: 1 means *pick* and 0 means *don't pick*. Fix the positions from left to right as elements 1, 2, and 3. Then the path to $\{1, 2\}$ is the bitmask 110: pick 1, pick 2, don't pick 3.

Bitmask for elements 1, 2, 3	Subset
000	$\emptyset$
001	$\{3\}$
010	$\{2\}$
011	$\{2, 3\}$
100	$\{1\}$
101	$\{1, 3\}$
110	$\{1, 2\}$
111	$\{1, 2, 3\}$

Every subset determines exactly one path in the tree and exactly one bitmask. Conversely, every path or bitmask specifies exactly one subset. Thus listing the power set is the same task as listing all n -bit patterns. The element order only fixes the bitmask positions; it does not give an order to the set itself.

There are two choices for each of the n positions, so

$$|\mathcal{P}(A)| = 2^n.$$

5.3 Simple Recursive Pseudocode

The arguments are the two parts of a tree node: `picked` and `remaining`. Each recursive call makes one decision.

```
Subsets(picked, remaining):
    if remaining is empty:
        print picked
        return

    x = an element of remaining
    rest = remaining without x
    Subsets(picked union {x}, rest)    // pick x
    Subsets(picked, rest)              // don't pick x

Subsets(empty set, {1, 2, 3})
```

Here union and difference form new sets. Every subset follows exactly one sequence of decisions, and every call removes one remaining element, so the procedure terminates and prints every subset once.

5.4 Simple Bitmask Pseudocode

Count from 0 to $2^n - 1$ and use each number's binary representation as a bitmask. In this pseudocode, `binary(number)` returns that representation as n bits, adding leading zeros when needed. For $n = 3$, `binary(2)` is 010.

```
A = [1, 2, 3]
n = length(A)

for number = 0 to 2^n - 1:
    bitmask = binary(number)
    subset = empty set
    for i = 0 to n - 1:
        if bitmask[i] == 1:
            subset = subset union {A[i]}
    print subset
```

The bit positions follow the order of `A`, just as in the table. This loop and the recursive procedure list the same subsets.

6 The Cardinality of Infinite Sets

Two sets have the same cardinality when their elements can be **paired**: every element on each side has exactly one partner, with nothing left out and no element used twice.

An infinite set is **countably infinite** if we can list its elements as $b_0, b_1, b_2, \dots$, with every element appearing at a finite position, exactly once. Pairing i with b_i shows that its cardinality equals $|\mathbb{N}|$. A set is **countable** if it is finite or countably infinite; an infinite set that cannot be listed this way is **uncountable**.

6.1 The Even Natural Numbers

Let $E = \{0, 2, 4, 6, \dots\}$. Pair each natural number n with $2n$:

$\mathbb{N}$	0	1	2	3	4	...
E	0	2	4	6	8	...

Claim 4. $|E| = |\mathbb{N}|$.

Proof. Different natural numbers have different partners: $2n = 2m$ implies $n = m$. Every even natural number is $2k$ for a unique $k \in \mathbb{N}$, so every element of E is paired exactly once. $\square$

Although $E \subset \mathbb{N}$, the two sets have the same cardinality. An infinite set can have the same size as a proper subset.

6.2 The Integers

Alternate positive and negative integers so that neither side waits for an infinite list to finish:

$\mathbb{N}$	0	1	2	3	4	5	6	...
$\mathbb{Z}$	0	1	-1	2	-2	3	-3	...

Claim 5. $|\mathbb{Z}| = |\mathbb{N}|$.

Proof. Put 0 at position 0. For each $k \geq 1$, put k at position $2k - 1$ and $-k$ at position $2k$. Every positive position is uniquely odd or even, so each receives one entry. Every integer appears at exactly one finite position: 0 at 0, k at $2k - 1$, and $-k$ at $2k$. $\square$

6.3 The Reals: Cantor's Diagonal Argument

Can we also list all real numbers? Cantor's idea was to show that if such a mapping existed, he could construct a number that was not on the list which would be a contradiction.

Claim 6. *The interval $(0, 1)$ is uncountable.*

Proof by contradiction. Suppose all real numbers in $(0, 1)$ appear in a list $r_1, r_2, r_3, \dots$. Write $r_i = 0.d_{i1}d_{i2}d_{i3}\dots$. When a number has two decimal expansions, use the one that does not end in repeating 9s; for example, use $0.5000\dots$ instead of $0.4999\dots$.

Construct $y = 0.c_1c_2c_3 \dots$ by changing each diagonal digit:

$$c_i = \begin{cases} 2, & \text{if } d_{ii} = 1, \\ 1, & \text{otherwise.} \end{cases}$$

The picture illustrates the construction on a hypothetical list.

digit position		1	2	3	4	5	6	
r_1	0.	1	4	1	5	9	2	$\dots$
r_2	0.	2	7	1	8	2	8	$\dots$
r_3	0.	6	1	2	0	3	4	$\dots$
r_4	0.	9	9	9	1	0	0	$\dots$
r_5	0.	0	5	7	8	4	6	$\dots$
r_6	0.	3	3	3	3	3	1	$\dots$
$\vdots$								$\ddots$
y	0.	2	1	1	2	1	2	$\dots$



change each
diagonal digit

Since y uses only the digits 1 and 2, it lies in $(0, 1)$ and has no ambiguous decimal expansion. For every i , its i th digit differs from the i th digit of r_i , so $y \neq r_i$. Thus y is missing from the supposedly complete list, a contradiction. $\square$

The real numbers are therefore uncountable too: a listing of all of $\mathbb{R}$ would give a listing of $(0, 1)$ by keeping just the entries in that interval.

6.4 Larger Infinities: A Glimpse Beyond the Course

Cantor's theorem says that the power set of any set has strictly greater cardinality:

$$|A| < |\mathcal{P}(A)|.$$

This holds even for infinite sets. Repeatedly taking power sets therefore gives ever larger infinities; there is no largest size of set.

We write $\aleph_0$ (“aleph-null”) for the size of $\mathbb{N}$. The symbols $\aleph_1, \aleph_2, \dots$ denote successively larger infinite cardinalities. The size of $\mathbb{R}$ is called the **continuum**; we should not simply identify it with $\aleph_1$. The general theorem and this hierarchy are beyond the scope of our class.

7 Russell's Paradox and the Set of All Sets

Sets can contain other sets. It is therefore natural to ask whether we can collect *all sets* into one set U . Every set would then be an element of U , including any new set we form from its elements.

The difficulty appears when we use the predicate $S \notin S$ to select the sets that do not contain themselves. From the proposed U , form

$$R = \{S \in U : S \notin S\}.$$

This uses the same predicate-based construction as earlier in the lecture. If U existed, R would be a set, and therefore $R \in U$. We must then be able to decide whether R itself belongs to R .

7.1 Both Membership Choices Lead to a Contradiction

The diagram follows both possible answers. In each branch, the defining condition for R forces the opposite of the answer we started with.

